Transformations

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

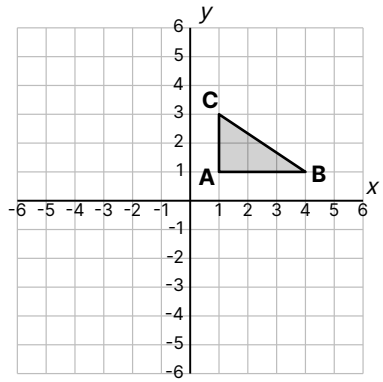
KEY WORDS translation *(nekehanga)* · reflection *(whakaata)* · rotation *(hurihanga)* · enlargement *(whakarahi)*

A Translation ●○○

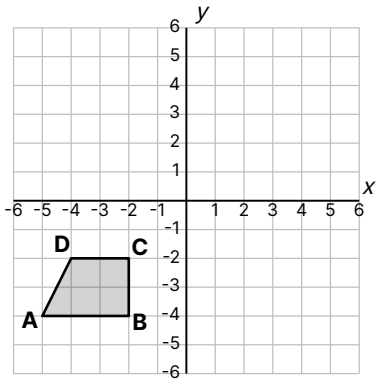
Remember: A translation slides every point the same distance in the same direction. Label the image A′, B′, C′.

1. Draw the image of each shape.

a) Translate the shape 5 units left and 2 units up.

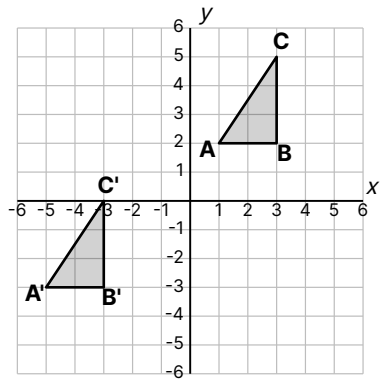


b) Translate the shape 6 units right and 5 units up.



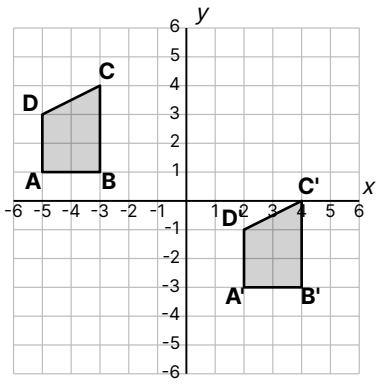
2. Each shape ABC has been translated to A′B′C′. Describe the translation.

a)



_____

b)



_____

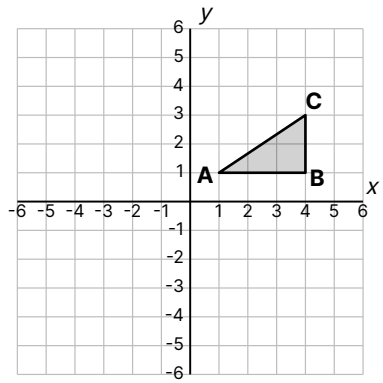
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B Reflection ●●○

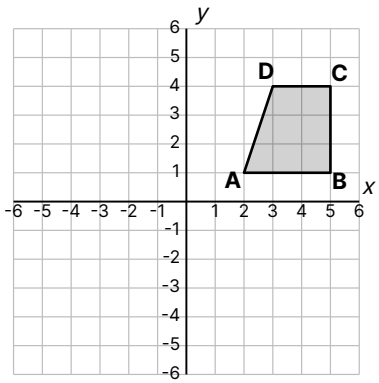
Remember: Each point and its image are the same distance from the mirror line, on opposite sides of it.

3. Draw the image of each shape.

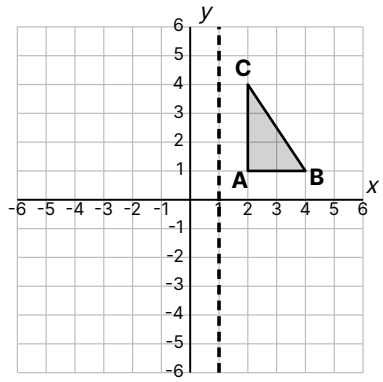
a) Reflect the shape in the x-axis.



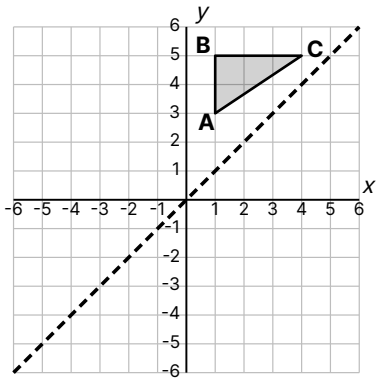
b) Reflect the shape in the y-axis.



c) Reflect the shape in the line $x = 1$ (the dashed line).



d) Reflect the shape in the line $y = x$ (the dashed line).

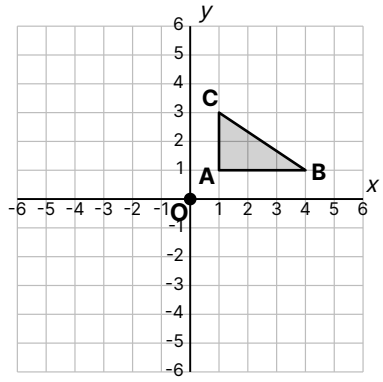


C Rotation ●●●

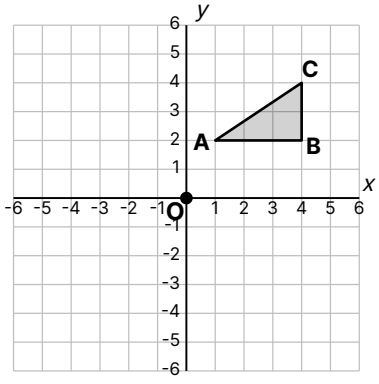
Remember: Every point turns through the same angle about the centre. Tracing paper helps: trace the shape, hold the centre still and turn.

4. Draw the image of each shape.

a) Rotate the shape 180° about the point O.

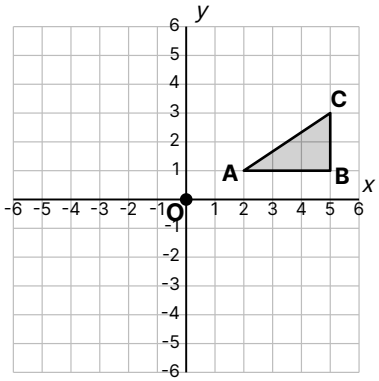


b) Rotate the shape 90° clockwise about the point O.

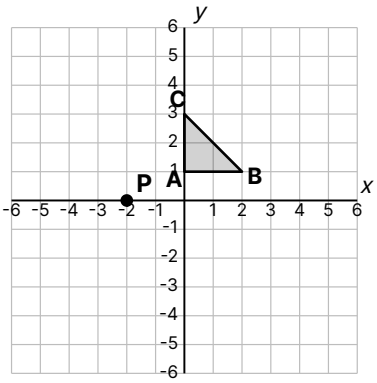


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c) Rotate the shape 90° anticlockwise about the point O.



d) Rotate the shape 180° about the point P.

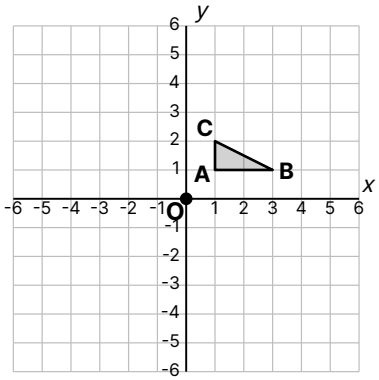


D Enlargement ●●●

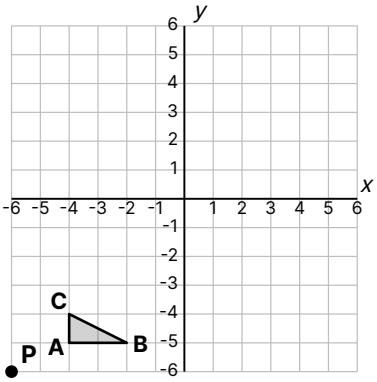
Remember: Multiply the distance from the centre to each point by the scale factor.

5. Draw the image of each shape.

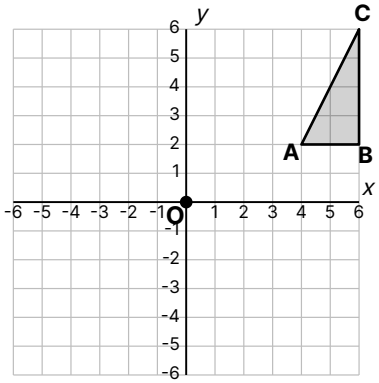
a) Enlarge the shape by a scale factor of 2, with centre O.



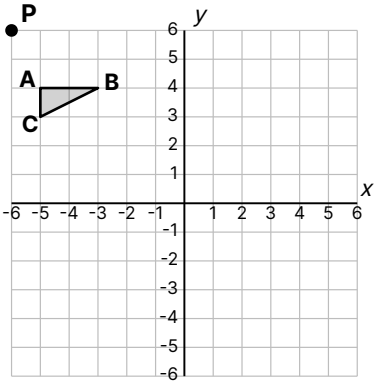
b) Enlarge the shape by a scale factor of 3, with centre P.



c) Enlarge the shape by a scale factor of 0.5, with centre O.



d) Enlarge the shape by a scale factor of 2, with centre P.



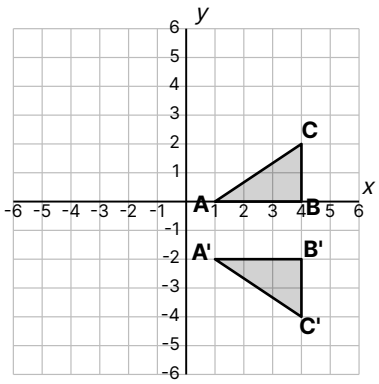
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E Naming the transformation ●●●

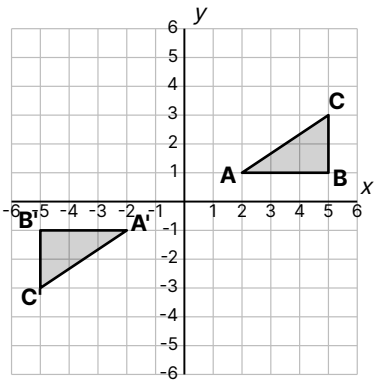
⚠ **Watch out:** A full description needs every detail: the mirror line, or the centre and angle, or the centre and scale factor.

6. Shape ABC has been transformed to A′B′C′. Describe the single transformation fully.

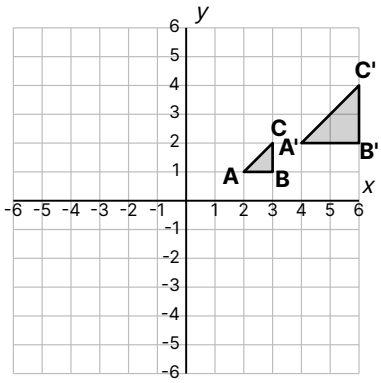
a)



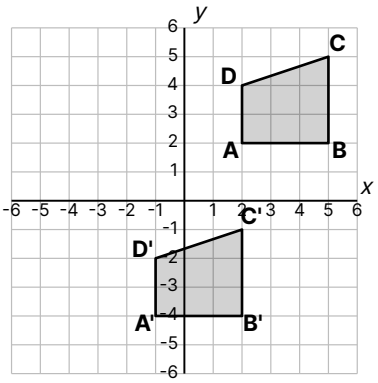
b)



c)



d)



Extension

7. Without drawing, write the coordinates of the image of the point $P(3,\ -2)$ after each transformation.

a) Reflection in the x-axis

b) Reflection in the line $y = x$

c) Rotation of 90° clockwise about (0, 0)

d) Rotation of 180° about (0, 0)

e) Enlargement, scale factor 3, centre (0, 0)

f) Reflection in the y-axis, then a translation 2 right and 3 up

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Transformations

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

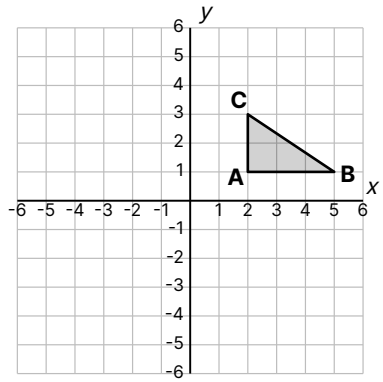
KEY WORDS translation *(nekehanga)* · reflection *(whakaata)* · rotation *(hurihanga)* · enlargement *(whakarahi)*

A Translation ●○○

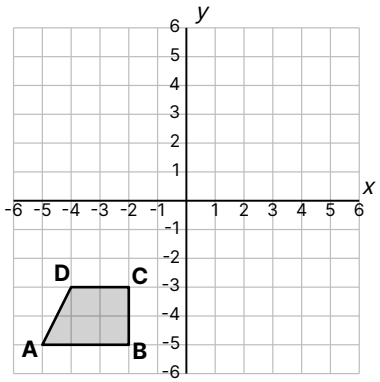
Remember: A translation slides every point the same distance in the same direction. Label the image A′, B′, C′.

1. Draw the image of each shape.

a) Translate the shape 6 units left and 2 units up.

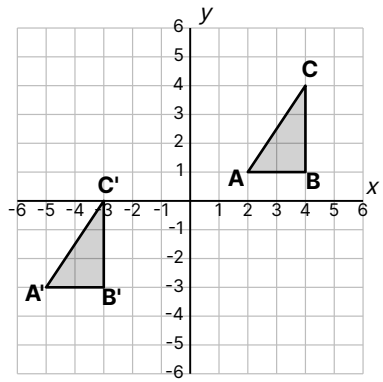


b) Translate the shape 5 units right and 6 units up.



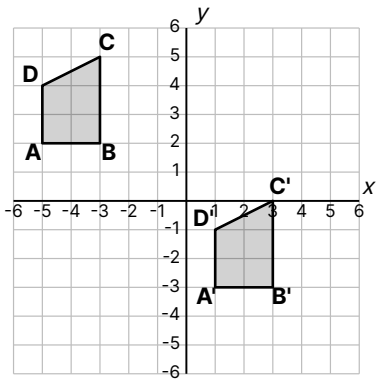
2. Each shape ABC has been translated to A′B′C′. Describe the translation.

a)



_____

b)



_____

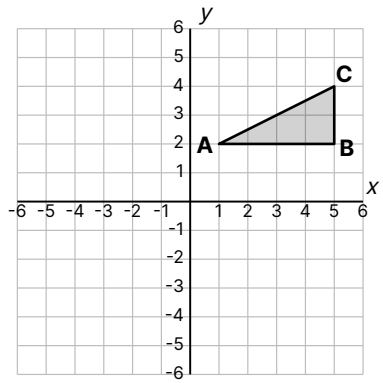
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B Reflection ●●○

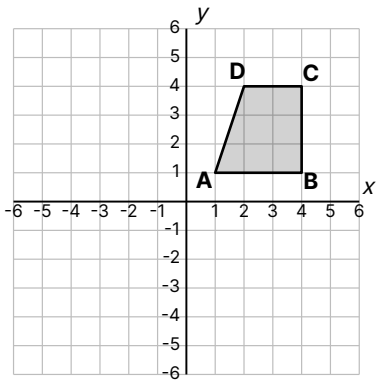
Remember: Each point and its image are the same distance from the mirror line, on opposite sides of it.

3. Draw the image of each shape.

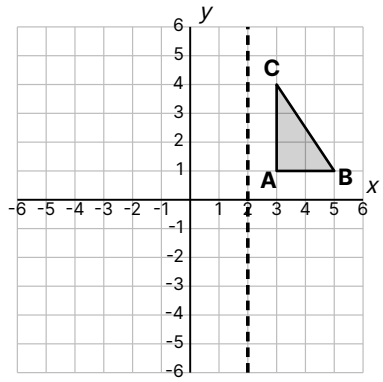
a) Reflect the shape in the x-axis.



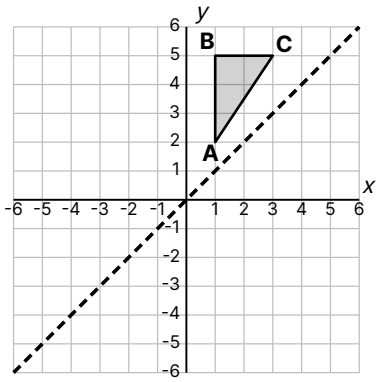
b) Reflect the shape in the y-axis.



c) Reflect the shape in the line $x = 2$ (the dashed line).



d) Reflect the shape in the line $y = x$ (the dashed line).

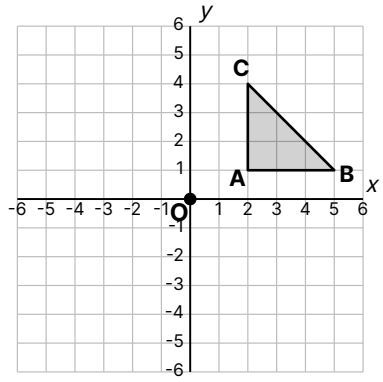


C Rotation ●●●

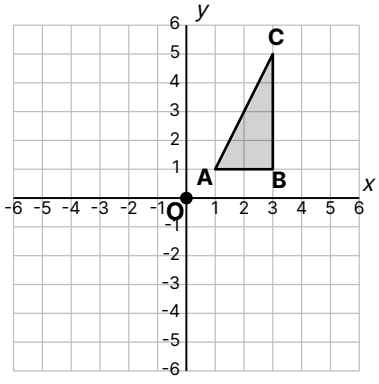
Remember: Every point turns through the same angle about the centre. Tracing paper helps: trace the shape, hold the centre still and turn.

4. Draw the image of each shape.

a) Rotate the shape 180° about the point O.

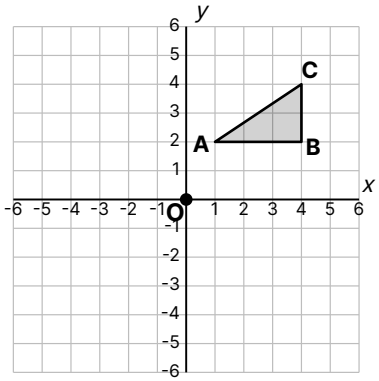


b) Rotate the shape 90° clockwise about the point O.

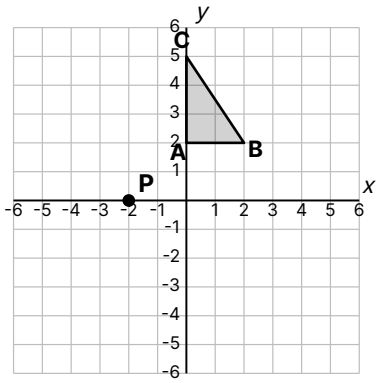


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c) Rotate the shape 90° anticlockwise about the point O.



d) Rotate the shape 180° about the point P.

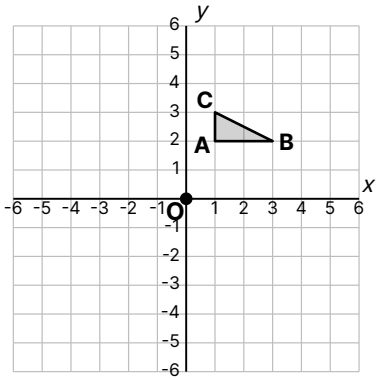


D Enlargement ●●●

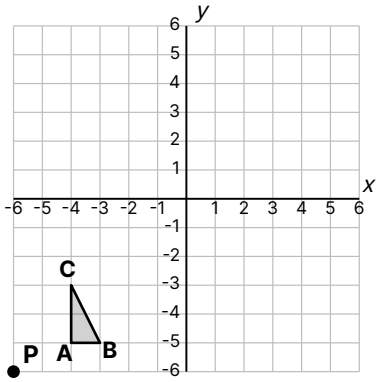
Remember: Multiply the distance from the centre to each point by the scale factor.

5. Draw the image of each shape.

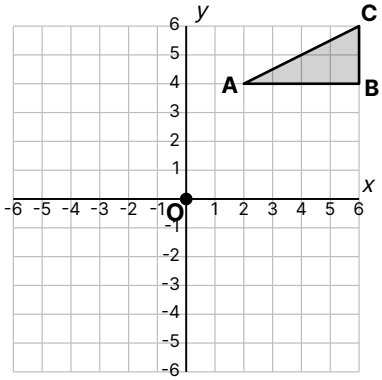
a) Enlarge the shape by a scale factor of 2, with centre O.



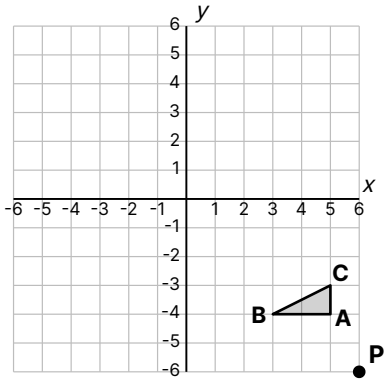
b) Enlarge the shape by a scale factor of 3, with centre P.



c) Enlarge the shape by a scale factor of 0.5, with centre O.



d) Enlarge the shape by a scale factor of 2, with centre P.



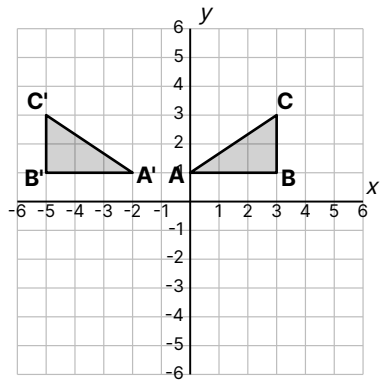
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E Naming the transformation ●●●

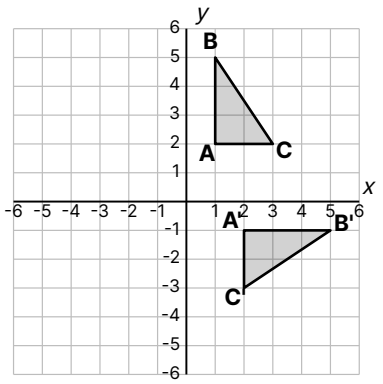
⚠ **Watch out:** A full description needs every detail: the mirror line, or the centre and angle, or the centre and scale factor.

6. Shape ABC has been transformed to A′B′C′. Describe the single transformation fully.

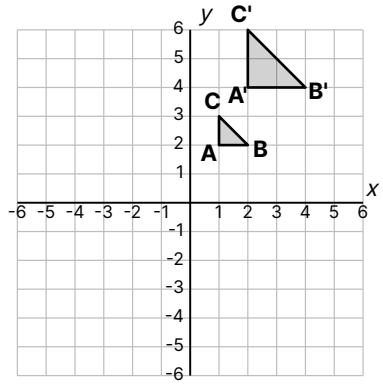
a)



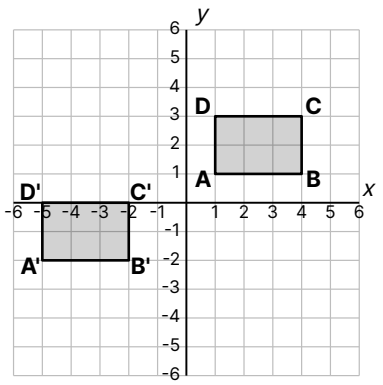
b)



c)



d)



★ Extension

7. Without drawing, write the coordinates of the image of the point $P(-4,\ 1)$ after each transformation.

a) Reflection in the x-axis

b) Reflection in the line $y = x$

c) Rotation of 90° clockwise about (0, 0)

d) Rotation of 180° about (0, 0)

e) Enlargement, scale factor 3, centre (0, 0)

f) Reflection in the y-axis, then a translation 2 right and 3 up

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